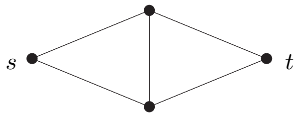
zero, by letting $f_2(\vec{e}) := \bar{1} \in \mathbb{Z}_2$ for all edges e of H. Setting f_2 to zero on e_1 too then turns it into a $\mathbb{Z}_2$-circulation on G. Now $f := f_2 \times f_3$ is a $(\mathbb{Z}_2 \times \mathbb{Z}_3)$-flow on G with f_2 zero on all the edges at u, as desired. □

Exercises

1.$^-$ Prove Proposition 6.2.1 by induction on $|S|$.

2. (i)$^-$ Given $n \in \mathbb{N}$, find a capacity function for the network below such that the algorithm from the proof of the max-flow min-cut theorem will need more than n augmenting paths W if these are badly chosen.



(ii)$^+$ Show that, if all augmenting paths are chosen as short as possible, their number is bounded by a function of the size of the network.

3.$^+$ Derive Menger's Theorem 3.3.5 from the max-flow min-cut theorem.

(Hint. The edge version is easy. For the vertex version, apply the edge version to a suitable auxiliary graph.)

4.$^-$ Let $f\colon \vec{E} \to H$ be a circulation on G and $g\colon H \to H'$ a group homomorphism. Show that $g \circ f$ is a circulation on G. Is $g \circ f$ an H'-flow if f is an H-flow?

5. View the group of circulations on a graph with values in $\mathbb{Z}_2$ as a vector space over $\mathbb{Z}_2$. Find a space in Chapter 1.9 to which it is isomorphic, and write down an explicit isomorphism.

6. Let H be an abelian group, $G = (V, E)$ a connected graph, T a spanning tree, and f a map from the orientations of the edges in $E \setminus E(T)$ to H that satisfies (F1). Show that f extends uniquely to a circulation on G with values in H.

7. (continued)

Let $\mathcal{V}_H = \mathcal{V}_H(G)$ be the group of all maps $V \to H$, and $\mathcal{E}_H = \mathcal{E}_H(G)$ the group of all maps $\vec{E} \to H$ satisfying (F1), both with pointwise addition. Every $\varphi \in \mathcal{V}_H$ defines a $\psi \in \mathcal{E}_H$ by $\psi(e, x, y) := \varphi(y) - \varphi(x)$.

(i) Show that these ψ form a subgroup $\mathcal{B}_H = \mathcal{B}_H(G)$ of $\mathcal{E}_H$ with $\mathcal{B}_H = \{\, \psi \in \mathcal{E}_H \mid \psi(\vec{C}) = 0 \text{ for every oriented cycle } C \subseteq G \,\}$, where $\psi(\vec{C}) := \sum_{\vec{e} \in \vec{C}} \psi(\vec{e})$.

(ii) Show that every map $\vec{E}(T) \to H$ satisfying (F1) extends uniquely to a map in $\mathcal{B}_H$.

8.$^+$ (continued)

Let $\mathcal{C}_H$ denote the group of all circulations on G with values in H.

(i) Show that $\mathcal{E}_H/\mathcal{B}_H$ is isomorphic to $\mathcal{C}_H$.

(ii) Show that $\mathcal{E}_H/\mathcal{C}_H$ is isomorphic to $\mathcal{B}_H$.

9.$^-$ Given $k \geqslant 1$, show that a graph has a k-flow if and only if each of its blocks has a k-flow.

10.$^-$ Show that $\varphi(G/e) \leqslant \varphi(G)$ whenever G is a multigraph and e an edge of G. Does this imply that, for every k, the class of all multigraphs admitting a k-flow is closed under taking minors?

11.$^-$ Work out the flow number of K^4 directly, without using any results from the text.

12. Let G be a graph with a k-flow, where $k \geqslant 3$. Find a cubic graph that has a k-flow and from which G can be obtained by contracting edges and identifying nonadjacent vertices. Can you find a construction that does not require the assumption of $k \geqslant 3$?

Do not use the 6-flow Theorem 6.6.1 for the following three exercises.

13. Show that $\varphi(G) < \infty$ for every bridgeless multigraph G.

14. Let G be a bridgeless connected graph with n vertices and m edges. By considering a normal spanning tree of G, show that $\varphi(G) \leqslant m - n + 2$.

15. Assume that a graph G has m spanning trees such that no edge of G lies in all of these trees. Show that $\varphi(G) \leqslant 2^m$.

16. Show that every graph with a Hamilton cycle has a 4-flow. (A *Hamilton cycle* of G is a cycle in G that contains all the vertices of G.)

17. A family of (not necessarily distinct) subgraphs of a graph G is called a *double cover* of G if every edge of G lies on exactly two of these subgraphs. The *cycle double cover conjecture* asserts that every bridgeless multigraph admits a double cover by cycles. Prove the conjecture for graphs with a 4-flow.

18.$^-$ Determine the flow number of $C^5 * K^1$, the wheel with 5 spokes.

19. Find bridgeless graphs G and $H = G - e$ such that $2 < \varphi(G) < \varphi(H)$.

20. Prove Proposition 6.4.1 without using Theorem 6.3.3.

21.$^+$ Prove that a plane triangulation is 3-colourable if and only if all its vertices have even degree.

22. Show that the 3-flow conjecture for planar multigraphs is equivalent to Grötzsch's Theorem 5.1.3.

23. (i)$^-$ Show that the four colour theorem is equivalent to the non-existence of a planar snark, i.e. to the statement that every cubic bridgeless planar multigraph has a 4-flow.

(ii) Can 'bridgeless' in (i) be replaced by '3-connected'?

24.$^{+}$ Show that a graph $G = (V, E)$ has a k-flow if and only if it has an orientation D that directs, for every $X \subseteq V$, at least $1/k$ of the edges in $E(X, \overline{X})$ from X towards $\overline{X}$.

Notes

Network flow theory is an application of graph theory that has had a major and lasting impact on its development over decades. As is illustrated already by the fact that Menger's theorem can be deduced easily from the max-flow min-cut theorem (Exercise 3), the interaction between graphs and networks may go either way: while 'pure' results in areas such as connectivity, matching and random graphs have found applications in network flows, the intuitive power of the latter has boosted the development of proof techniques that have in turn brought about theoretic advances.

The classical reference for network flows is L.R. Ford & D.R. Fulkerson, *Flows in Networks*, Princeton University Press 1962. More recent and comprehensive accounts are given by R.K. Ahuja, T.L. Magnanti & J.B. Orlin, *Network flows*, Prentice-Hall 1993, by A. Frank in his chapter in the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995, and by A. Schrijver, *Combinatorial optimization*, Springer 2003. An introduction to graph algorithms in general is given in A. Gibbons, *Algorithmic Graph Theory*, Cambridge University Press 1985.

If one recasts the maximum flow problem in linear programming terms, one can derive the max-flow min-cut theorem from the linear programming duality theorem; see A. Schrijver, *Theory of integer and linear programming*, Wiley 1986.

The more algebraic theory of group-valued flows and k-flows has been developed largely by Tutte; he gives a thorough account in his monograph W.T. Tutte, *Graph Theory*, Addison-Wesley 1984. The fact that the number of k-flows of a multigraph is a polynomial in k, whose values can be bounded in terms of the corresponding values of the flow polynomial, was proved by M. Kochol, Polynomials associated with nowhere-zero[8] flows, *J. Comb. Theory, Ser. B* **84** (2002), 260–269.

Tutte's flow conjectures are covered also in F. Jaeger's survey, Nowhere-zero flow problems, in (L.W. Beineke & R.J. Wilson, eds.) *Selected Topics in Graph Theory 3*, Academic Press 1988. For the flow conjectures, see also T.R. Jensen & B. Toft, *Graph Coloring Problems*, Wiley 1995. The 6-flow theorem is due to P.D. Seymour, Nowhere-zero 6-flows, *J. Comb. Theory, Ser. B* **30** (1981), 130–135. This paper also indicates how Tutte's 5-flow conjecture reduces to snarks. Our proof is from M. Devos and K. Nurse, A short proof of Seymour's 6-flow theorem, arXiv:2307.04768. The proof of the 4-flow conjecture for cubic graphs announced in 1998 Robertson, Sanders, Seymour and Thomas has not yet been entirely written up. C. Thomassen, The weak 3-flow conjecture and the weak circular flow conjecture, *J. Comb. Theory, Ser. B* **102** (2012), 521–529, proved that every graph of large enough edge-connectivity k

[8] In the literature, the term 'flow' is often used to mean what we have called 'circulation', i.e. flows are not required to be nowhere zero unless this is stated explicitly.